

A 16nm, 1Mb, 1-to-8b-Configurable 444.21TOPS/W Fully Digital SRAM Compute-in-Memory Macro for Hybrid SNN-CNN Edge Computing

Publisher: **IEEE** [Cite This](#)

<< Results

[Yao-Kai Yeh](#) ; [Jian-Wei Su](#) ; [Ting-Hao Hsu](#) ; [Mai Tseng](#) ; [Jen-Chun Tien](#) ; [Ko-Chi Chen](#) 

805
Full
Text Views



Down
PDF

Abstract:

We present a silicon-verified fully digital SRAM-CIM macro that supports multi-bit hybrid SNN-CNN processing. Key innovations include: (1) a compact CFD-SC-RC architectur... [Show More](#)

▼ **Metadata**

Recommended for You (Beta)

A 15T SRAM Cell-Based Fully-Digital Computing-in-Memory Macro Supporting High Parallelism and Fine...

A 22 nm Floating-Point ReRAM Compute-in-Memory Macro Using...

An Integer-Floating-Point Dual-Mode Gain-Cell Computing-in-Memory...

[Learn More](#)

 Contents

Spiking neural networks (SNNs) [1] have demonstrated superior energy efficiency in event-driven applications such as event-based vision and audio processing, while convolutional neural networks (CNNs) [2]– [3] provide high accuracy in image and voice recognition tasks at the edge. Hybrid SNN-CNN models [4]– [6] have shown promise in balancing accuracy and compute energy, enabling robust edge inference for various applications. Meanwhile, SRAM-based digital compute-in-memory (SRAM-DCIM) [7] has been widely explored to reduce data movement and improve energy efficiency, while outperforming analog SRAM-CIM (SRAM-ACIM) [8] in MAC accuracy. Combining SRAM-DCIM with hybrid SNN-CNN models is a promising solution for diverse edge-AI tasks.

However, as shown in Fig. 30.8.1, designing SRAM-DCIM macros supporting hybrid SNN-CNN introduces two key challenges: (1) extra hardware overhead for supporting both SNN and CNN, and excessive compute energy in the digital domain; and (2) area-energy trade-off between partial sum (PSUM) storage and weight (W) update during input (IN) and W data mapping.

This work addresses these challenges with two key techniques: (1) a compact fully-digital SNN-CNN reconfigurable computing (CFD-SC-RC) architecture that shares SNN-CNN MAC circuit to reduce area and energy overhead, and supports adaptive model and precision selection across layers; and (2) a PSUM storage-weight reuse optimized data mapping (PS-WR-ODM) scheme that balances PSUM-storage minimization and weight-reuse maximization, achieving optimized register usage and weight-update energy. The test chip, fabricated in a foundry 16nm FinFET CMOS process, demonstrates 444.21 TOPS/W with a 1bIN-8bW-



14bOUT configuration in SNN mode, and 62.84TOPS/W with an 8bIN-8bW-22bOUT configuration in CNN mode.

[Figure 30.8.2](#) illustrates the proposed 1Mb CFD-SC-RC SRAM-DCIM macro, consisting of 16 64kb banks. Each bank integrates a control block, I/O block, IN/W scheduler, WL driver, 32 reconfigurable inference blocks (RIBs), an adder tree, and a digital multi-membrane potential dynamics behavior (MPDB) unit (DM-MPDBU). Each RIB contains a 2kb SRAM array, a weight stationary cell (WSC), eight cross-model dual-mode local computing cells (CMDM-LCCs), and a shift-and-add (SaA) circuit. Naive SRAM-ACIM based hybrid SNN-CNN macros are vulnerable to process variations in MPDB functions, while prior SRAM-DCIM based hybrid SNN-CNN macros [\[4\]](#) support only a single MPDB type, limiting flexibility for diverse model requirements. Both architectures lack hardware sharing, incurring large area overhead. The proposed CFD-SC-RC architecture shares SNN-CNN MAC circuits to reduce area overhead, provides high tolerance to process variation, and, via the DM-MPDBU, additionally supports multiple MPDB functions beyond prior works [\[4\]](#), including integrate-and-fire (IF), leaky integrate-and-fire (LIF), and integer quadratic integrate-and-fire (IQIF) [\[9\]](#). The IF mode provides simple spike generation; the LIF mode incorporates leakage dynamics for temporal noise filtering; and the IQIF mode introduces nonlinear integration for improved accuracy, enabling general-purpose SNN operations at the edge.

[Figure 30.8.3](#) illustrates the CFD-SC-RC computation flow using the proposed CMDM-LCC and DM-MPDBU. Conventional bit-wise (BW)-IN multipliers (BWMs) for SNN cannot support word-wise (WW)-IN multiplication for CNN. To enable both BW-IN MAC for SNN and configurable WW-IN MAC for CNN, the proposed CMDM-LCC integrates two MUX-based multipliers (MULs), an INT8 toggle reducer (TR), dual-mode weight mapping (INT4/INT8) in the array, and an SaA circuit. Within one bank, each of eight CMDM-LCCs performs MAC operations with INs and Ws (W_{0-3}). In INT4 mode, the TR is disabled and four 4b MACs are executed per cycle: W_{0-1} map to two 4b weights ($A[3:0]$, $B[3:0]$) and W_{2-3} to another two ($C[3:0]$, $o[3:0]$). In INT8 mode, the TR eliminates redundant toggles, enabling two 8b MACs per cycle: W_{0-1} correspond to the four LSBs of $A[7:0]$ and $B[7:0]$, while W_{2-3} correspond to their four MSBs. The partial MACs ($pMAC_{0-7}$) are sent directly to the adder tree in SNN mode, while in CNN mode $pMAC_{0-7}$ are passed to the SaA circuit for WW-accumulation before entering the adder tree. Meanwhile, conventional MPDB blocks only support a single MPDB function, limiting flexibility for diverse SNN models. The proposed DM-MPDBU supports three functions: (1) IF mode-membrane potential (MP) accumulates until reaching threshold (peak), then fires and resets; (2) LIF mode-MP accumulates with leakage subtraction before proceeding as IF; and (3) IQIF mode-MP accumulates with an integer quadratic (IQ) term dynamically defined by parameters ($pdeth$, $rest$, $thresh$, α , β): if $MP[t] < pdeth$, $IQ = (\alpha \times rest - MP[t]) > 3$; otherwise, $IQ = (\beta \times (MP[t] - thresh)) > 3$.

[Figure 30.8.4](#) illustrates the proposed PS-WR-ODM scheme. In conventional timestep (TS)-first data mapping, all input neurons (IN_{0-K}) are accessed sequentially within each TS, minimizing PSUM storage but incurring high weight-update energy and long latency because weight reuse cannot be leveraged for early MPDB operations. In contrast, prior weight-reuse data mapping [\[10\]](#) accesses same-index INs (e.g., IN_0) across all TSs (TS_{0-N}), maximizing weight reuse and throughput but requiring large PSUM storage. The proposed PS-WR-ODM adopts a hybrid strategy in which TSs are grouped into blocks with user-defined size (e.g., pairs of adjacent TSs), and weight-reuse mapping is applied within each block. This approach achieves balanced performance in weight-update energy and latency with only marginal PSUM buffer overhead. For example, as shown in the timing diagrams, the conventional TS-first data mapping incurs long weight-load cycles, while the previous weight-reuse mapping demands excessive PSUM buffering. The proposed PS-WR-ODM achieves comparable latency reduction and energy savings without significant PSUM overhead.

[Figure 30.8.5](#) shows the simulated performance of the proposed schemes. The CFD-SC-RC reduces macro area by 33% and improves energy efficiency by $2.31 \times$ over prior hybrid SNN-CNN CIM design [\[4\]](#). The PS-WR-ODM achieves $> 122 \times$ improvement in FoM1, defined as: energy efficiency/(area $\times$ latency), compared to conventional timestep-first and weight-reuse-first [\[10\]](#) mappings. The combined implementation of CFD-SC-RC, CMDM-LCC, and PS-WR-ODM further delivers an FoM2, defined as inference accuracy $\times$ energy efficiency/(area $\times$ latency), of $5.48 \times$ over baseline designs [\[4\]](#), [\[10\]](#).

[Figure 30.8.6](#) presents the measurement results of the 16nm SRAM-DCIM macro and the position chart versus prior silicon-verified works. The Shmoo plot confirms 1.3ns computing latency (T_{MAC}) at 0.8V with an 8bIN-8bW-22bOUT configuration. The proposed macro achieves up to $8.2 \times$ improvement in FoM3, defined as: IN precision $\times$ W precision $\times$ energy efficiency, over prior works. [Figure 30.8.7](#) presents the die

micrograph and chip performance summary. To the best of our knowledge, this is this chip is the first silicon-verified fully digital SRAM-CIM macro that supports multi-bit hybrid SNN-CNN processing.

ACKNOWLEDGEMENT

The authors would like to thank NTHU-ITRI collaboration and NSTC for their financial and manufacturing support.



Figure 30.8.1:
Motivation and challenges for hybrid SNN-CNN SRAM-CIM.



Figure 30.8.2:
Structure of proposed SRAM-DCIM macro and CFD-SC-RC operation.



Figure 30.8.3:
CMDM-ICC and DM-MPDBU structure and operation.



Figure 30.8.4:
PS-WR-ODM operational concept.



Figure 30.8.5:
Simulated performance of proposed schemes.



Figure 30.8.6:
Measurement results and position chart.



Figure 30.8.7:
Die micrograph and performance summary table.

[VIEW PURCHASED
DOCUMENTS](#)

[PROFESSION AND
EDUCATION](#)

[WORLDWIDE: +1 732
981 0060](#)

[TECHNICAL INTERESTS](#)

[CONTACT & SUPPORT](#)

[About IEEE *Xplore*](#) | [Contact Us](#) | [Help](#) | [Accessibility](#) | [Terms of Use](#) | [Nondiscrimination Policy](#) | [IEEE Ethics Reporting](#)  | [Sitemap](#) | [IEEE Privacy Policy](#)

A public charity, IEEE is the world's largest technical professional organization dedicated to advancing technology for the benefit of humanity.

© Copyright 2026 IEEE - All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies.